Quantum Noise Simulation

Assessment of noise models, channels, simulation and mitigation

QUESTION PAPER

Time allowed:	3 hours	**Maximum marks:**	240
Questions:	80 (4 sections × 20)	**Answer key:**	separate Answer Book

Name of the candidate: _____ **Roll No.:** _____

General instructions

1. This question paper has **four sections** of **20 questions** each: **A** Mathematical and Quantum Foundations, **B** Quantum Noise Fundamentals, **C** Noise Models and Simulation, **D** Advanced Noise Analysis and Implementation.
2. Every question is of the **single-correct** type: four options, exactly one correct. Question formats used: single-correct MCQ, matching lists (List-I/List-II), Assertion–Reason, Statement I/II and paragraph-based questions.
3. For every question choose **ONE** option, (A), (B), (C) or (D), and record it against the question number in the answer script provided by the invigilator.
4. Calculators are **not** allowed. Rough work may be done on separate sheets.
5. Unless a question says otherwise, use the **conventions** printed below; they apply to every section.
6. The detailed solutions are in the separate *Answer Book*; do not consult it before finishing the test.

Marking scheme (every question)

Each question has **four** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

Full marks:	+3 if ONLY the correct option is chosen;
Zero marks:	0 if none of the options is chosen (question unanswered);
Negative marks:	−1 in all other cases (wrong option, or more than one option chosen).

Conventions and notation used throughout the paper

- Qubit 0 is written first: $|q_0 q_1 \ldots\rangle$; two-qubit matrices use the basis order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. X, Y, Z are the Pauli matrices, H the Hadamard gate, $R_P(\theta) = e^{-i\theta P/2}$; CNOT flips the target when the control is 1.
- A qubit state is $\rho = \frac{1}{2}(I + r_x X + r_y Y + r_z Z)$ with Bloch vector $\vec{r}$, $|\vec{r}| \le 1$; purity $\mathrm{Tr}\,\rho^2$. State fidelity with a pure target: $F = \langle\psi|\rho|\psi\rangle$. Measurement probabilities are the diagonal of ρ; $\langle Z\rangle = P(0) - P(1)$.
- A channel is $\mathcal{E}(\rho) = \sum_k K_k \rho K_k^\dagger$ with $\sum_k K_k^\dagger K_k = I$. Bit flip: $\sqrt{1-p}\,I, \sqrt{p}\,X$; phase flip: $\sqrt{1-p}\,I, \sqrt{p}\,Z$; bit-phase flip: $\sqrt{1-p}\,I, \sqrt{p}\,Y$; depolarizing (CUDA-Q): $\sqrt{1-p}\,I$, $\sqrt{p/3}\,X$, $\sqrt{p/3}\,Y$, $\sqrt{p/3}\,Z$ (two-qubit: $\sqrt{1-p}\,II$ and the 15 other Paulis with $p/15$); amplitude damping: $\mathrm{diag}(1, \sqrt{1-\gamma})$, $\sqrt{\gamma}\,|0\rangle\langle 1|$; phase damping: $\mathrm{diag}(1, \sqrt{1-\lambda})$, $\mathrm{diag}(0, \sqrt{\lambda})$.
- T_1: $\rho_{11}(t) = \rho_{11}(0)e^{-t/T_1}$; T_2: $|\rho_{01}(t)| = |\rho_{01}(0)|e^{-t/T_2}$; $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\varphi}$. ln is the natural logarithm.
- **Notebook** (`Noise_Simulation.ipynb`, defaults): 15-qubit GHZ circuit (H on qubit 0, CNOT chain), the chosen channel with probability p after every `h`, `x`, `ry`, `rz` gate and a two-qubit channel after every CNOT; no measurement noise; 10 000 shots, seed 42. **Previously recorded run**: CUDA-Q 0.16.0, CPU target `stim`, no GPU, depolarizing $p = 0.01$: population 0.8936, X-basis parity 0.6910, $(1-p)^G = 0.8601$. No physical quantum processor was used.

SECTION A · Mathematical and Quantum Foundations Q.1–20

Syllabus: probability and error statistics, complex numbers and phases, matrices and linear operators, density matrices, pure and mixed states, purity, eigenvalues, partial trace, the Bloch sphere and fidelity of states.
Formats: Single-correct MCQ (Q.1–14); Matching lists (Q.15–16); Assertion–Reason (Q.17); Statement I / II (Q.18); Paragraph based (Q.19–20). Each question: +3 / 0 / −1.

Q.1 The density matrix of the pure state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$ is

(A) $\frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$ **(B)** $\frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}$ **(C)** $\frac{1}{2}\begin{pmatrix}1 & -i\\ i & 1\end{pmatrix}$ **(D)** $\frac{1}{2}\begin{pmatrix}1 & i\\ -i & 1\end{pmatrix}$

Q.2 The purity $\operatorname{Tr}\rho^2$ of $\rho = \begin{pmatrix}0.7 & 0.3\\ 0.3 & 0.3\end{pmatrix}$ is

(A) 0.58 **(B)** 1 **(C)** 0.76 **(D)** 0.52

Q.3 The Bloch vector (r_x, r_y, r_z) of $\rho = \begin{pmatrix}0.8 & 0.2 - 0.1i\\ 0.2 + 0.1i & 0.2\end{pmatrix}$ is

(A) $(0.2,\ 0.1,\ 0.6)$ **(B)** $(0.4,\ 0.2,\ 0.6)$
(C) $(0.4,\ -0.2,\ 0.6)$ **(D)** $(0.4,\ 0.2,\ 0.8)$

Q.4 Which of the following matrices is *not* a valid density matrix?

(A) $\begin{pmatrix}0.5 & 0.5\\ 0.5 & 0.5\end{pmatrix}$ **(B)** $\begin{pmatrix}0.5 & -0.3i\\ 0.3i & 0.5\end{pmatrix}$ **(C)** $\begin{pmatrix}0.9 & 0\\ 0 & 0.1\end{pmatrix}$ **(D)** $\begin{pmatrix}0.6 & 0.5\\ 0.5 & 0.4\end{pmatrix}$

Q.5 The operators $K_0 = \begin{pmatrix}1 & 0\\ 0 & 0.8\end{pmatrix}$ and $K_1 = \begin{pmatrix}0 & b\\ 0 & 0\end{pmatrix}$ $(b \geq 0)$ form a trace-preserving set of Kraus operators exactly when b equals

(A) 0.6 **(B)** 0.36 **(C)** 0.8 **(D)** 0.4

Q.6 A qubit is prepared in $\frac{1}{\sqrt{2}}(|0\rangle + e^{i\varphi}\,|1\rangle)$ where the phase φ is random and uniformly distributed in $[-\frac{\pi}{2}, \frac{\pi}{2}]$. The average value of $\langle X\rangle$ is

(A) $\pi/4 \approx 0.79$ **(B)** $1/2$ **(C)** $2/\pi \approx 0.64$ **(D)** 0

Q.7 A circuit has 20 gates; each fails independently with probability $p = 0.01$. The probability that *at least one* gate fails is

(A) 0.182 **(B)** 0.818 **(C)** 0.200 **(D)** 0.165

Q.8 An outcome has true probability 0.9. It is estimated from 10 000 shots. The standard error of the estimate is

(A) 0.03 **(B)** 0.003 **(C)** 0.0009 **(D)** 0.009

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Q.9 An X gate is implemented as $R_X(\pi+\varepsilon)$ instead of $R_X(\pi)$, with $\varepsilon = 0.1$ rad. Applied to $|0\rangle$, the probability of measuring 0 (an error) is closest to

(A) 0.0025 **(B)** 0.01 **(C)** 0.1 **(D)** 0.05

Q.10 The fidelity $F = \langle +|\rho|+\rangle$ between the target $|+\rangle$ and the mixed state $\rho = 0.8\,|+\rangle\langle +| + 0.2\,\frac{I}{2}$ is

(A) 0.6 **(B)** 0.9 **(C)** 0.8 **(D)** 0.95

Q.11 For the two-qubit state $|\psi\rangle = \sqrt{0.8}\,|00\rangle + \sqrt{0.2}\,|11\rangle$, the purity $\mathrm{Tr}\,\rho_A^2$ of the reduced state of qubit A is

(A) 0.8 **(B)** 0.5 **(C)** 0.68 **(D)** 1

Q.12 The larger eigenvalue of $\rho = \begin{pmatrix} 0.7 & 0.3 \\ 0.3 & 0.3 \end{pmatrix}$ is closest to

(A) 0.700 **(B)** 0.861 **(C)** 1.000 **(D)** 0.760

Q.13 Two independent qubits are in the states $\rho_A = \mathrm{diag}(0.9, 0.1)$ and $\rho_B = \mathrm{diag}(0.8, 0.2)$. In the state $\rho_A \otimes \rho_B$, the probability that both qubits give the *same* outcome is

(A) 0.74 **(B)** 0.50 **(C)** 0.26 **(D)** 0.72

Q.14 A qubit with Bloch vector (r_x, r_y, r_z) is acted on by $\rho \mapsto X\rho X$. Its new Bloch vector is

(A) $(-r_x,\ -r_y,\ -r_z)$ **(B)** $(-r_x,\ r_y,\ r_z)$ **(C)** $(r_x,\ -r_y,\ -r_z)$ **(D)** $(r_x,\ r_y,\ -r_z)$

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Q.15 Match each state (List-I) with its Bloch vector (List-II).

List-I		List-II	
(P)	$\|1\rangle$	**(1)**	$(0,\ 0,\ 0.5)$
(Q)	$\frac{I}{2}$	**(2)**	$(0,\ 0,\ 0)$
(R)	$\frac{1}{2}\left(\begin{smallmatrix} 1 & -i \\ i & 1 \end{smallmatrix}\right)$	**(3)**	$(0,\ 0,\ -1)$
(S)	$\mathrm{diag}(0.75,\ 0.25)$	**(4)**	$(0,\ 1,\ 0)$

(A) P→3, Q→1, R→4, S→2 **(B)** P→3, Q→2, R→1, S→4
(C) P→1, Q→2, R→4, S→3 **(D)** P→3, Q→2, R→4, S→1

Q.16 Match each matrix (List-I) with its property (List-II).

List-I		List-II	
(P)	$\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$	**(1)**	unitary but not Hermitian
(Q)	$\frac{1}{2}\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$	**(2)**	Hermitian and unitary
(R)	$\left(\begin{smallmatrix} 1 & 0 \\ 0 & i \end{smallmatrix}\right)$	**(3)**	neither Hermitian nor unitary (a Kraus operator of amplitude damping, up to a factor)
(S)	$\left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$	**(4)**	a rank-1 projector and a valid density matrix

(A) P→4, Q→2, R→1, S→3 **(B)** P→2, Q→4, R→3, S→1
(C) P→1, Q→4, R→2, S→3 **(D)** P→2, Q→4, R→1, S→3

Q.17 **Assertion (A):** Every mixed single-qubit state lies strictly inside the Bloch ball ($|\vec{r}| < 1$).
Reason (R): For $\rho = \frac{1}{2}(I + \vec{r}\cdot\vec{\sigma})$ one has $\mathrm{Tr}\,\rho^2 = \frac{1+|\vec{r}|^2}{2}$, which equals 1 only when $|\vec{r}| = 1$.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

Q.18 **Statement I:** Every 2×2 Hermitian matrix with trace 1 is a valid density matrix.
Statement II: The diagonal entries of a density matrix in the computational basis are the probabilities of the outcomes of a computational-basis measurement.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Paragraph for Questions 19 and 20

A qubit should have been prepared in $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. State tomography of the real (noisy) preparation gives the density matrix

$$\rho = \begin{pmatrix} 0.55 & 0.35 \\ 0.35 & 0.45 \end{pmatrix}.$$

Q.19 The Bloch vector and the purity of the prepared state are

(A) $(0.7,\ 0,\ 0.1)$ and 0.50
(B) $(0.7,\ 0,\ 0.55)$ and 0.85
(C) $(0.35,\ 0,\ 0.1)$ and 0.75
(D) $(0.7,\ 0,\ 0.1)$ and 0.75

Q.20 The fidelity $\langle +|\rho|+\rangle$ of the preparation, and the probability that a computational-basis measurement gives 1, are

(A) 0.70 and 0.45 **(B)** 0.85 and 0.45 **(C)** 0.92 and 0.45 **(D)** 0.85 and 0.55

SECTION B · Quantum Noise Fundamentals Q.21–40

Syllabus: decoherence, bit flips, phase flips, bit-phase flips, depolarizing noise, amplitude and phase damping, T_1, T_2 and pure dephasing, measurement (readout) errors, coherent versus stochastic errors.
Formats: Single-correct MCQ (Q.21–34); Matching lists (Q.35–36); Assertion–Reason (Q.37); Statement I / II (Q.38); Paragraph based (Q.39–40). Each question: +3 / 0 / −1.

Q.21 Which single-qubit state is left *unchanged* by the bit-flip channel $\rho \mapsto (1-p)\rho + pX\rho X$ for every p?

(A) $\frac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle)$ **(B)** $|1\rangle$
(C) $|0\rangle$ **(D)** $|+\rangle$

Q.22 A phase-flip channel with $p = 0.1$ acts on $|+\rangle$. The state is then measured in the X basis. The probability of the outcome "+" is

(A) 0.95 **(B)** 1.0 **(C)** 0.8 **(D)** 0.9

Q.23 A depolarizing channel $\rho \mapsto (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z)$ with $p = 0.03$ acts on $|0\rangle$. The probability of then measuring 1 is

(A) 0.03 **(B)** 0.04 **(C)** 0.02 **(D)** 0.01

Q.24 Amplitude damping with $\gamma = 0.36$ acts on $|+\rangle$. The resulting density matrix is

(A) $\begin{pmatrix} 0.68 & 0.40 \\ 0.40 & 0.32 \end{pmatrix}$ **(B)** $\begin{pmatrix} 0.68 & 0.32 \\ 0.32 & 0.32 \end{pmatrix}$ **(C)** $\begin{pmatrix} 0.50 & 0.40 \\ 0.40 & 0.50 \end{pmatrix}$ **(D)** $\begin{pmatrix} 0.68 & 0.50 \\ 0.50 & 0.32 \end{pmatrix}$

Q.25 Phase damping with parameter λ multiplies the off-diagonal elements by $\sqrt{1-\lambda}$; a phase-flip channel multiplies them by $1-2p$. The phase-flip probability p that has the same effect as phase damping with $\lambda = 0.36$ is

(A) 0.18 **(B)** 0.36 **(C)** 0.10 **(D)** 0.20

Q.26 A qubit with $T_1 = 50\,\mu$s starts in $|1\rangle$. After what time is $P(1) = \frac{1}{2}$?

(A) 50 μs **(B)** 69.3 μs **(C)** 25 μs **(D)** 34.7 μs

Q.27 A qubit has $T_1 = 100\,\mu$s and pure-dephasing time $T_\varphi = 200\,\mu$s. Using $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\varphi}$, T_2 equals

(A) 150 μs **(B)** 66.7 μs **(C)** 200 μs **(D)** 100 μs

Q.28 Which pair (T_1, T_2) is *physically impossible* in the standard model?

(A) $T_1 = 100\,\mu$s, $T_2 = 30\,\mu$s **(B)** $T_1 = 50\,\mu$s, $T_2 = 120\,\mu$s
(C) $T_1 = 50\,\mu$s, $T_2 = 100\,\mu$s **(D)** $T_1 = 80\,\mu$s, $T_2 = 80\,\mu$s

Q.29 Gates take 50 ns and the qubit has $T_2 = 100\,\mu$s. Requiring the coherence factor e^{-t/T_2} to stay at least 0.9, the largest number of sequential gates is about

(A) 210 **(B)** 21 **(C)** 2000 **(D)** 105

Q.30 A detector reads 1 when the qubit is in $|0\rangle$ with probability 0.02, and reads 0 when it is in $|1\rangle$ with probability 0.05. For a perfectly prepared $|+\rangle$, the probability of reading 1 is

(A) 0.515 **(B)** 0.475 **(C)** 0.485 **(D)** 0.500

Q.31 A perfect 3-qubit GHZ state is measured with the readout errors of the previous question ($P(1|0) = 0.02$, $P(0|1) = 0.05$, independent per qubit). The probability of observing 000 or 111 is about

(A) 0.899 **(B)** 0.950 **(C)** 0.857 **(D)** 0.941

Q.32 Which of the following is a *coherent* (unitary) error rather than decoherence or a measurement error?

(A) the discriminator labelling a $|1\rangle$ outcome as 0

(B) a systematic over-rotation of every X gate caused by a miscalibrated pulse amplitude

(C) a random phase that differs from shot to shot and averages to dephasing

(D) energy relaxation of $|1\rangle$ to $|0\rangle$ (a T_1 process)

Q.33 After a CNOT, CUDA-Q's `Depolarization2(p)` applies each of the 15 non-identity two-qubit Paulis with probability $p/15$. For $p = 0.015$, the probability that the outcome of a Z-basis measurement of the control qubit is flipped by this error is

(A) 0.010 **(B)** 0.008 **(C)** 0.0075 **(D)** 0.015

Q.34 A 15-qubit GHZ state suffers only phase-flip noise. Which measurement reveals the errors?

(A) the parity $\langle X^{\otimes 15}\rangle$ measured after a Hadamard on every qubit

(B) the probability $P(0\cdots0)$ alone

(C) the number of distinct outcomes in the Z basis

(D) the Z-basis population $P(0\cdots0) + P(1\cdots1)$

Q.35 Match each channel (List-I) with a set of Kraus operators (List-II).

List-I		List-II	
(P)	bit flip	**(1)**	$\mathrm{diag}(1, \sqrt{1-\lambda})$, $\mathrm{diag}(0, \sqrt{\lambda})$
(Q)	phase flip	**(2)**	$\sqrt{1-p}\,I$, $\sqrt{p}\,Z$
(R)	amplitude damping	**(3)**	$\left(\begin{smallmatrix}1 & 0\\ 0 & \sqrt{1-\gamma}\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}0 & \sqrt{\gamma}\\ 0 & 0\end{smallmatrix}\right)$
(S)	phase damping	**(4)**	$\sqrt{1-p}\,I$, $\sqrt{p}\,X$

(A) P→4, Q→1, R→3, S→2 **(B)** P→4, Q→2, R→3, S→1

(C) P→2, Q→4, R→3, S→1 **(D)** P→4, Q→2, R→1, S→3

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Q.36 Match each channel (List-I) with its action on the Bloch vector (r_x, r_y, r_z) (List-II).

List-I		**List-II**	
(P)	depolarizing (CUDA-Q convention, p)	**(1)**	r_x, r_y scaled, r_z unchanged
(Q)	amplitude damping (γ)	**(2)**	all three components multiplied by the same factor $1 - \frac{4p}{3}$
(R)	phase damping (λ)	**(3)**	r_x unchanged, r_y and r_z multiplied by $1 - 2p$
(S)	bit flip (p)	**(4)**	r_x, r_y multiplied by $\sqrt{1-\gamma}$, $r_z \mapsto \gamma + (1-\gamma) r_z$

(A) P→2, Q→1, R→4, S→3 **(B)** P→2, Q→4, R→3, S→1

(C) P→3, Q→4, R→1, S→2 **(D)** P→2, Q→4, R→1, S→3

Q.37 **Assertion (A):** Amplitude damping is not unital: it does not map the maximally mixed state $I/2$ to itself.

Reason (R): Amplitude damping multiplies the off-diagonal elements of ρ by $\sqrt{1-\gamma}$.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A

(B) Both A and R are true, but R is *not* the correct explanation of A

(C) A is true, but R is false

(D) A is false, but R is true

Q.38 **Statement I:** The phase-flip channel with $p = \frac{1}{2}$ maps every single-qubit state to a state with zero off-diagonal elements.

Statement II: The phase-flip channel with $p = 1$ destroys all quantum information.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

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Paragraph for Questions 39 and 40

A superconducting qubit has $T_1 = 80\,\mu$s and $T_2 = 60\,\mu$s. It is prepared in $|+\rangle$ and then idles for $t = 20\,\mu$s. Use the standard model: populations relax towards $|0\rangle$ as $\rho_{11}(t) = \rho_{11}(0)e^{-t/T_1}$, and the coherence decays as $|\rho_{01}(t)| = |\rho_{01}(0)|e^{-t/T_2}$, with $\frac{1}{T_2} = \frac{1}{2T_1} + \frac{1}{T_\varphi}$ (T_φ = pure-dephasing time).

Q.39 After the idle period, ρ_{11} and $|\rho_{01}|$ are approximately

(A) 0.389 and 0.358 **(B)** 0.500 and 0.358 **(C)** 0.389 and 0.316 **(D)** 0.389 and 0.389

Q.40 The fidelity $\langle+|\rho|+\rangle$ after the idle period, and the pure-dephasing time T_φ, are

(A) 0.889 and 96 μs

(B) 0.716 and 96 μs

(C) 0.858 and 96 μs

(D) 0.858 and 43.6 μs

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SECTION C · Noise Models and Simulation Q.41–60

Syllabus: Kraus operators, quantum channels and CPTP maps, density-matrix evolution, noisy circuits, sampling, gate noise, simulator targets and memory, figures of merit, and the code and recorded results of `Noise_Simulation.ipynb`.
Formats: Single-correct MCQ (Q.41–54); Matching lists (Q.55–56); Assertion–Reason (Q.57); Statement I / II (Q.58); Paragraph based (Q.59–60). Each question: +3 / 0 / −1.

Q.41 On the target `density-matrix-cpu` a kernel prepares $|0\rangle$ and applies `rx(t, q[0])`, with `t = 0.0` passed as a kernel argument. The noise model attaches `cudaq.DepolarizationChannel(0.3)` to every `rx`. `cudaq.observe` of Z returns

(A) 0.7 **(B)** 0.6 **(C)** 0.4 **(D)** 1.0

Q.42 The same experiment is repeated, but the kernel now contains `rx(0.0, q[0])` with the angle written as a *literal constant*. With CUDA-Q 0.16 the result is $\langle Z\rangle = 1.0$. The best explanation is

(A) the compiler removes the constant identity rotation, so there is no `rx` gate left to carry the noise channel
(B) noise channels are applied only before a gate, never after it
(C) `density-matrix-cpu` ignores noise models
(D) the depolarizing channel acts only on non-zero rotation angles by definition

Q.43 Which set of operators defines a valid (completely positive, trace-preserving) quantum channel?

(A) $\left\{\left(\begin{smallmatrix}1 & 0\\ 0 & 0.6\end{smallmatrix}\right),\ \left(\begin{smallmatrix}0 & 0.6\\ 0 & 0\end{smallmatrix}\right)\right\}$
(B) $\{I,\ \sqrt{0.1}\,X\}$
(C) $\{\sqrt{0.5}\,I,\ \sqrt{0.5}\,X,\ \sqrt{0.5}\,Z\}$
(D) $\{\sqrt{0.7}\,I,\ \sqrt{0.3}\,Y\}$

Q.44 The bit-flip channel with $p = 0.2$ acts on $\rho = \begin{pmatrix}0.9 & 0.3\\ 0.3 & 0.1\end{pmatrix}$. The output is

(A) $\begin{pmatrix}0.74 & 0.18\\ 0.18 & 0.26\end{pmatrix}$ **(B)** $\begin{pmatrix}0.74 & 0.30\\ 0.30 & 0.26\end{pmatrix}$ **(C)** $\begin{pmatrix}0.82 & 0.30\\ 0.30 & 0.18\end{pmatrix}$ **(D)** $\begin{pmatrix}0.90 & 0.18\\ 0.18 & 0.10\end{pmatrix}$

Q.45 Two depolarizing channels (CUDA-Q convention) with $p_1 = 0.03$ and $p_2 = 0.06$ act one after the other. They equal a single depolarizing channel with p equal to

(A) 0.1168 **(B)** 0.0882 **(C)** 0.0876 **(D)** 0.0900

Q.46 In `Noise_Simulation.ipynb` the 15-qubit GHZ circuit has G noisy gate locations (1 Hadamard and 14 CNOTs, each followed by a depolarizing channel). The notebook's no-error probability $(1-p)^G$ at $p = 0.02$ is

(A) 0.7000 **(B)** 0.9800 **(C)** 0.7386 **(D)** 0.5455

Q.47 For `NOISE_MODEL = "bit_flip"`, the notebook's CNOT channel is `Pauli2` with probabilities $p(1-p)$ for IX and XI and p^2 for XX. The no-error probability of the 15-qubit GHZ circuit at $p = 0.01$ is

(A) 0.7472 **(B)** 0.8601 **(C)** 0.7397 **(D)** 0.9801

Q.48 The density-matrix method stores 4^N complex numbers of 16 bytes. The memory for $N = 13$ qubits is about

(A) 1.07 GB **(B)** 131 kB **(C)** 268 MB **(D)** 4.29 GB

Q.49 Which simulation can the stabilizer target `stim` perform *correctly*?

(A) a GHZ circuit with amplitude-damping noise

(B) a GHZ circuit followed by a T gate, with bit-flip noise

(C) a GHZ circuit (H, CNOT) with depolarizing noise after every gate

(D) the notebook's mirror circuit with random R_Y, R_Z rotations and depolarizing noise

Q.50 On the density-matrix path the notebook calls `as_counts_from_rho(rho, n, shots)`. What does this function do?

(A) it takes the diagonal of ρ and returns it unchanged as counts, without sampling

(B) it takes the diagonal of the exact noisy ρ as outcome probabilities, reorders the indices so that qubit 0 is written first, and draws `shots` counts from a multinomial distribution

(C) it measures the density-matrix simulator `shots` times with `cudaq.sample`

(D) it rounds every entry of ρ to the nearest multiple of 1/`shots`

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Q.51 The ideal distribution of a 2-qubit circuit is $P(00) = P(11) = 0.5$. The noisy distribution is $Q(00) = Q(11) = 0.45$, $Q(01) = Q(10) = 0.05$. The Hellinger fidelity $F_H = \left(\sum_x \sqrt{P_x Q_x}\right)^2$ and the total variation distance $\frac{1}{2}\sum_x |P_x - Q_x|$ are

(A) 0.900 and 0.200 **(B)** 0.949 and 0.100 **(C)** 0.810 and 0.100 **(D)** 0.900 and 0.100

Q.52 A 2-qubit GHZ state is measured after Hadamards on both qubits (X basis): counts `{00: 450, 11: 400, 01: 80, 10: 70}`. The parity $\langle X \otimes X \rangle$ is

(A) 0.40 **(B)** 0.15 **(C)** 0.70 **(D)** 0.85

Q.53 For $N = 20$ qubits, by what factor does the memory of exact density-matrix simulation exceed that of one trajectory (state vector) in the same precision?

(A) 20 **(B)** about 10^{12} **(C)** 2 **(D)** about 10^6

Q.54 In `build_noise_model` the line `model.add_all_qubit_channel("x", two, num_controls=1)` means

(A) the channel `two` is applied once, at the end of the circuit, to the first two qubits

(B) the channel `two` replaces the CNOT gate

(C) after every single-qubit `x` gate, the channel `two` acts on that qubit and its neighbour

(D) after every CNOT (an `x` gate with one control), the two-qubit channel `two` acts on its control and target qubits

Q.55 Match each CUDA-Q simulation target (List-I) with its property for noisy simulation (List-II).

List-I		List-II	
(P)	`stim`	**(1)**	Monte-Carlo trajectories on a GPU, memory $\propto 2^N$, statistical
(Q)	`density-matrix-cpu`	**(2)**	noiseless state vector: not offered by the noise notebook (it ignores noise models)
(R)	`nvidia` with a noise model	**(3)**	Clifford circuits with Pauli noise, polynomial cost, hundreds of qubits
(S)	`qpp-cpu`	**(4)**	exact ρ, any Kraus channel, memory $16 \cdot 4^N$ bytes

(A) P→3, Q→4, R→1, S→2
(B) P→3, Q→4, R→2, S→1
(C) P→4, Q→3, R→1, S→2
(D) P→3, Q→1, R→4, S→2

Q.56 Match each figure of merit of the notebook (List-I) with what it is sensitive to or requires (List-II).

List-I		List-II	
(P)	Z-basis GHZ population	**(1)**	needs the full density matrix; also the off-diagonal (coherence) elements
(Q)	X-basis parity $\langle X^{\otimes N}\rangle$	**(2)**	bit flips (errors that change computational-basis outcomes)
(R)	state fidelity $\langle\psi\|\rho\|\psi\rangle$	**(3)**	phase errors, which are invisible in the Z basis
(S)	purity $\mathrm{Tr}\,\rho^2$	**(4)**	mixedness of the state, whatever the target

(A) P→2, Q→3, R→4, S→1
(B) P→2, Q→1, R→3, S→4
(C) P→2, Q→3, R→1, S→4
(D) P→3, Q→2, R→1, S→4

Q.57 **Assertion (A):** For depolarizing noise, the GHZ state fidelity $\langle\mathrm{GHZ}|\rho|\mathrm{GHZ}\rangle$ is *exactly* equal to the no-error probability $(1-p)^G$.

Reason (R): For Pauli noise, a run without any error produces the ideal state, and no error occurs with probability $(1-p)^G$.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

Q.58 **Statement I:** In CUDA-Q 0.16 the density-matrix path of the notebook calls `set_noise` before `get_state`, because `get_state` has no `noise_model` argument.

Statement II: The density-matrix method needs $16 \cdot 4^N$ bytes, so $N = 14$ needs about 4.3 GB.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true, but Statement II is false
(D) Statement I is false, but Statement II is true

Paragraph for Questions 59 and 60

The previously recorded run of `Noise_Simulation.ipynb` simulates the 15-qubit GHZ circuit (H on qubit 0 and a chain of 14 CNOTs) with depolarizing noise after every gate on the stabilizer target `stim` (10 000 shots, seed 42). Part of its table:

p	population $P(0\cdots0)+P(1\cdots1)$	parity $\langle X^{\otimes 15}\rangle$	$(1-p)^G$	distinct outcomes
0.01	0.8936	0.6910	0.8601	96
0.1	0.3232	0.0160	0.2059	1165

Q.59 The probability mass found *outside* the two ideal GHZ outcomes at $p = 0.01$ and at $p = 0.1$ is

(A) 0.1399 and 0.7941 **(B)** 0.1064 and 0.6768

(C) 0.3090 and 0.9840 **(D)** 0.8936 and 0.3232

Q.60 Why is the population at $p = 0.01$ (0.8936) *larger* than $(1-p)^G = 0.8601$, while the parity (0.6910) is smaller?

(A) $(1-p)^G$ is an upper bound on the population, so the population must lie below it — the table contains a mistake

(B) $(1-p)^G$ counts every error, but Z-type errors leave the Z-basis population intact; the parity is also sensitive to those phase errors (and to Hadamard noise in the X-basis circuit)

(C) the stabilizer simulator under-reports errors because it cannot simulate depolarizing noise

(D) 10 000 shots are too few: with more shots the population would fall to exactly 0.8601

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SECTION D · Advanced Noise Analysis and Implementation Q.61–80

Syllabus: process and average gate fidelity, error rates and randomized benchmarking, zero-noise extrapolation, readout-error mitigation, dynamical decoupling, quantum error correction, twirling, CUDA-Q implementation and practical limitations.
Formats: Single-correct MCQ (Q.61–74); Matching lists (Q.75–76); Assertion–Reason (Q.77); Statement I / II (Q.78); Paragraph based (Q.79–80). Each question: +3 / 0 / −1.

Q.61 For a single-qubit depolarizing channel (CUDA-Q convention: Kraus operators $\sqrt{1-p}\,I$, $\sqrt{p/3}\,X, Y, Z$) the process (entanglement) fidelity is $F_{\text{pro}} = 1 - p$. Using $F_{\text{avg}} = \frac{dF_{\text{pro}}+1}{d+1}$, the average gate fidelity for $p = 0.003$ is

(A) 0.9980 **(B)** 0.9990 **(C)** 0.9970 **(D)** 0.99775

Q.62 Single-qubit randomized benchmarking fits the survival probability to $A\alpha^m + B$ (m = sequence length) and finds $\alpha = 0.99$. The error per Clifford $r = \frac{(d-1)(1-\alpha)}{d}$ is

(A) 0.010 **(B)** 0.0067 **(C)** 0.0025 **(D)** 0.005

Q.63 Zero-noise extrapolation measures $E(1) = 0.80$ at the native noise and $E(2) = 0.64$ at doubled noise. The *linear* extrapolation to $\lambda = 0$ gives

(A) 0.88 **(B)** 1.00 **(C)** 0.80 **(D)** 0.96

Q.64 ZNE with three noise levels gives $E(1) = 0.80$, $E(2) = 0.65$, $E(3) = 0.53$. The quadratic (Richardson) extrapolation $E(0) = 3E(1) - 3E(2) + E(3)$ equals

(A) 0.95 **(B)** 0.985 **(C)** 0.98 **(D)** 0.53

Q.65 A circuit has 10 gates of equal error. To amplify noise, 4 of them are "folded" ($G \to GG^\dagger G$), the others are left unchanged. Assuming the noise grows with the number of gates, the effective noise scale factor is

(A) 2.2 **(B)** 1.4 **(C)** 1.8 **(D)** 3.0

Q.66 Readout calibration of one qubit gives $P(\text{read } 0|0) = 0.98$ and $P(\text{read } 0|1) = 0.05$. An experiment reads 0 with frequency 0.60. The mitigated probability of the true state $|0\rangle$ is about

(A) 0.604 **(B)** 0.591 **(C)** 0.600 **(D)** 0.612

Q.67 Which statement about readout-error mitigation by inverting the calibrated confusion matrix A is correct?

(A) it reduces the number of shots needed, because A^{-1} removes the shot noise

(B) it requires quantum error correction to estimate A

(C) it also corrects gate errors that happened before the measurement

(D) $A^{-1}\mathbf{q}$ can contain small negative "quasi-probabilities", and the full $2^N \times 2^N$ matrix becomes impractical for many qubits

Q.68 A qubit in $|+\rangle$ idles for t with a random but *static* detuning $\delta \sim \mathcal{N}(0, \sigma^2)$, $\sigma = 1$ rad/μs, so it acquires the phase δt. At $t = 2\,\mu$s the average coherence $\langle X \rangle$ without and with a Hahn echo (X pulse at $t/2$ and again at the end) is

(A) 0.135 and 0.135 **(B)** 0.368 and 1.0 **(C)** 0.607 and 1.0 **(D)** 0.135 and 1.0

Q.69 Now the dephasing is Markovian: an independent phase-flip channel with $p = 0.02$ acts after each of 20 idle steps on $|+\rangle$. With a Hahn echo inserted at step 10, $\langle X \rangle$ at the end is

(A) 0 **(B)** 1.000
(C) 0.442, the same as without echo **(D)** 0.668

Q.70 In the 3-qubit bit-flip code each qubit flips independently with probability p, and encoding and correction are perfect. The logical error probability at $p = 0.1$, and the value of p above which encoding is worse than an unencoded qubit, are

(A) 0.028 and $p = 0.5$ **(B)** 0.010 and $p = 0.5$
(C) 0.028 and $p = 1/3$ **(D)** 0.030 and $p = 0.5$

Q.71 A surface-code memory has logical error rate $p_L \approx A(p/p_{\text{th}})^{(d+1)/2}$ for code distance d. With $p/p_{\text{th}} = 0.1$, increasing the distance from $d = 3$ to $d = 5$ reduces p_L by a factor of

(A) none: p_L increases by a factor of 10 **(B)** 10
(C) 2 **(D)** 100

Q.72 Each of $n = 60$ idle gates over-rotates coherently by $\varepsilon = 0.05$ rad about x. Compare with a stochastic bit flip of the same per-gate error probability $p = \sin^2(\varepsilon/2)$. Starting from $|0\rangle$, the probabilities of finding 1 after 60 gates are approximately

(A) 0.995 (coherent) and 0.036 (stochastic) **(B)** 0.0375 and 0.036
(C) 0.995 and 0.995 **(D)** 0.036 and 0.036

Q.73 Which statement about Pauli twirling (randomized compiling) is correct?

(A) it removes all errors, so the circuit becomes noiseless

(B) averaging a gate over random Pauli frames that cancel ideally turns its coherent error into a stochastic Pauli channel with the same average fidelity

(C) it requires a density-matrix simulator and cannot be done on hardware

(D) it converts stochastic Pauli noise into coherent errors to make them easier to correct

Q.74 The noise notebook runs on a CPU (no GPU visible) with the settings `CIRCUIT = "ghz"`, `NOISE_MODEL = "amplitude_damping"`, `SIMULATION_METHOD = "auto"`. Which method does it choose?

(A) `stabilizer` (target `stim`)

(B) `trajectory` (target `nvidia`)

(C) `statevector` (target `qpp-cpu`)

(D) `density_matrix` (target `density-matrix-cpu`)

Q.75 Match each technique (List-I) with its description (List-II).

List-I		List-II	
(P)	zero-noise extrapolation	**(1)**	invert a calibrated confusion matrix applied to the measured distribution
(Q)	readout-error mitigation	**(2)**	run the circuit at amplified noise levels and extrapolate the observable to zero noise
(R)	dynamical decoupling	**(3)**	encode logical qubits redundantly, measure syndromes and correct; needs physical errors below a threshold
(S)	quantum error correction	**(4)**	apply pulse sequences during idle periods to refocus slow (quasi-static) dephasing

(A) P→2, Q→1, R→4, S→3 **(B)** P→1, Q→2, R→4, S→3
(C) P→2, Q→1, R→3, S→4 **(D)** P→2, Q→4, R→1, S→3

Q.76 Match each physical cause (List-I) with its category (List-II).

List-I		List-II	
(P)	a pulse amplitude 1 % too large on every X gate	**(1)**	decoherence (non-unital, T_1)
(Q)	a random Pauli error after each gate	**(2)**	classical measurement (readout) error
(R)	energy loss of $\|1\rangle$ to the environment	**(3)**	coherent error (unitary, systematic)
(S)	the discriminator assigns some $\|1\rangle$ shots to outcome 0	**(4)**	stochastic (incoherent) gate noise

(A) P→4, Q→3, R→1, S→2 **(B)** P→3, Q→4, R→1, S→2
(C) P→3, Q→4, R→2, S→1 **(D)** P→3, Q→1, R→4, S→2

Q.77 **Assertion (A):** Zero-noise extrapolation increases the number of circuit executions needed to reach a given statistical precision.

Reason (R): ZNE corrects errors on the quantum state during the computation, in the same way as quantum error correction.

(A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A
(B) Both A and R are true, but R is *not* the correct explanation of A
(C) A is true, but R is false
(D) A is false, but R is true

Q.78 **Statement I:** `Noise_Simulation.ipynb` includes measurement (readout) errors in its noise model.

Statement II: The recorded results of `Noise_Simulation.ipynb` were obtained on a physical quantum processor.

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false

(C) Statement I is true, but Statement II is false

(D) Statement I is false, but Statement II is true

Paragraph for Questions 79 and 80

For the lecture (illustrative experiment, not part of the notebook), zero-noise extrapolation (ZNE) was applied to the parity $\langle X^{\otimes 4} \rangle$ of a 4-qubit GHZ state. The circuit was simulated exactly on `density-matrix-cpu` with the notebook's depolarizing noise model at the scaled probability $p = 0.01\lambda$ (noise scale factor λ). The ideal value is 1.

λ	1	2	3
$\langle X^{\otimes 4} \rangle$	0.9055	0.8189	0.7396

Q.79 The linear extrapolation from $\lambda = 1, 2$ and the Richardson (quadratic) extrapolation from $\lambda = 1, 2, 3$ give

(A) 0.9921 and 0.9994

(B) 0.9055 and 0.9921

(C) 0.9921 and 1.0021

(D) 0.9872 and 0.9994

Q.80 Which conclusion about this ZNE experiment is correct?

(A) on hardware the same result is obtained without scaling the noise, since λ is a software parameter

(B) ZNE has corrected the GHZ state itself, so a subsequent circuit could use the improved state

(C) the extrapolated values depend on the fitted model (an exponential fit gives $1.0021 > 1$, unphysical) and, with shot noise, would carry larger error bars than the raw value

(D) because the exact density matrix was used, ZNE is no longer needed: the raw value 0.9055 is already the ideal one

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